

I'll map this from “inside cabin camera” to the actual electronics stack, because the board depends a lot on whether you mean a prototype, dashcam-style device, or production ADAS/driver-monitoring system.

Thought for 5s

For an **inside camera of a car** — usually called **DMS/OMS**:

- **DMS** = Driver Monitoring System: face, eyes, gaze, drowsiness, distraction.
- **OMS** = Occupant Monitoring System: passengers, child presence, seat occupancy, gestures.

The image processing is usually done by one of these board/processor types:

1. Automotive vision SoC board

This is the common serious/prototype choice. The board has:

- Camera input: usually **MIPI CSI-2**, **GMSL**, or **FPD-Link**
- **ISP**: Image Signal Processor for raw camera image cleanup
- CPU: ARM cores
- AI accelerator: **NPU / DSP / GPU**
- Automotive interfaces: CAN, Ethernet, sometimes LIN
- Linux/QNX/RTOS support

Examples:

Board / SoC	Used for
TI TDA4VM / TDA4VM-Q1	ADAS, DMS, vision AI. Has 8 TOPS AI acceleration and vision processing hardware.
NVIDIA Jetson Orin / DRIVE AGX Orin	Heavy AI vision prototyping, multi-camera processing, autonomous/ADAS development. NVIDIA DRIVE AGX Orin is specifically for automotive development.
NXP i.MX 8M Plus / i.MX 9 family	Embedded vision, industrial/automotive-style camera processing. i.MX 8M Plus has integrated ISPs and long product availability.
Qualcomm Snapdragon Ride / automotive SoCs	Production automotive cockpit/ADAS platforms.

For a real car cabin camera system, this class is the “normal” answer.

2. Dedicated DMS/OMS ASIC

In production vehicles, companies often avoid a big development board and use a **dedicated camera AI processor**. This is smaller, lower power, and designed for one job: process cabin camera images and run AI models.

Examples:

Chip	Used for
OMNIVISION OAX8000	Dedicated DMS processor with integrated NPU + ISP + DDR3 for eye gaze / eye tracking.
OMNIVISION OAX4600	AI-enabled ASIC for combined DMS/OMS, with RGB-IR ISP, NPU, and embedded DDR3.
indie iND880 / iND881 family	Vision processor with ISP + AI/NPU for driver and occupant monitoring.

This is closer to what an OEM or Tier-1 supplier would use in a production vehicle. Tiny, efficient, but harder to develop with than Jetson/TI boards.

3. Dashcam / telematics camera board

If the goal is more like **recording video, fleet monitoring, or simple AI events**, the board is often a dashcam-style SoC board.

Typical chips:

- Ambarella CV series
- Novatek
- Hisilicon
- Rockchip
- Qualcomm/MediaTek embedded SoCs

These boards usually include:

- camera sensor input
- video encoder, often H.264/H.265
- storage
- LTE/Wi-Fi/GPS options
- maybe AI acceleration

Good for video recording + basic AI, but less “automotive safety-grade” than proper DMS/ADAS boards.

4. Simple prototype board

For early development, people often start with:

Board	Good for
Raspberry Pi 5 + camera	Cheap prototype, basic OpenCV
NVIDIA Jetson Orin Nano / NX	AI camera prototype, face/eye detection, TensorRT
Google Coral Dev Board / USB TPU	Lightweight TensorFlow Lite inference
Luxonis OAK-D	Camera module with onboard AI/depth processing

For your case, if you want to build/test an inside-car AI camera, I’d choose:

Prototype:

Jetson Orin Nano/NX + IR camera module

More automotive/professional:

TI TDA4VM-Q1 evaluation board + automotive camera

Production-style DMS/OMS:

OMNIVISION OAX4600/OAX8000 or indie vision processor

The key thing: the “board” is not just a normal microcontroller board. For car cabin image processing, you usually need a board with **ISP + AI accelerator + camera interface**, not just Arduino/ESP32.

I’ll price this as **development boards/kits you can actually buy**, because production ASICs like OAX8000/OAX4600 usually don’t have public unit pricing unless you contact the vendor or distributor.

Thought for 28s

Here are **rough current USD prices** for the boards/kits we discussed. Prices move a lot depending on distributor, stock, country, and whether it includes camera/power supply/case.

NVIDIA Jetson Orin Nano Super Developer Kit

\$559.99•Best AI prototype for advanced robotics.



Jetson Orin NX Super Developer Kit 16GB

\$1,149.00•Stronger Jetson for demanding AI projects.

NVIDIA Jetson AGX Orin Developer Kit

\$2,522.13•High-end AI computing power for complex tasks.

Luxonis OAK-D

\$329.00•Camera with integrated AI for vision projects.

Luxonis OAK-D W

\$479.00•Wide FOV camera for expansive vision needs.

Luxonis OAK-D CM4

\$429.00•Standalone camera host for flexible setups.



Raspberry Pi 5 Starter Kit

\$179.99 • Affordable starter kit for Raspberry Pi projects.

Seeed Studio Raspberry Pi 5 Starter Kit 8GB

\$339.94•8GB Raspberry Pi 5 kit with pre-loaded OS.

Board / Kit	Approx. price	Notes
Raspberry Pi 5 board only	about \$70–\$95	Cheap prototype board. Not strong for serious AI unless you add accelerator. Raspberry Pi’s newer listed prices put Pi 5 4GB at \$70 and 8GB at \$95.
Raspberry Pi 5 starter kit	about \$180–\$340	Includes accessories; useful if you do not already have power, SD, case, etc.
Google Coral Dev Board	about \$129.99	Has Edge TPU accelerator; good for TensorFlow Lite inference.
Luxonis OAK-D	about \$329	Camera + onboard AI/depth processing. Very convenient for vision prototyping.
Luxonis OAK-D W / PoE / CM4 variants	about \$429–\$579+	Better if you need wide FOV, Ethernet/PoE, or a built-in host module. Luxonis lists several OAK cameras from about \$169 to \$779 depending on model.
NVIDIA Jetson Orin Nano Super Developer Kit	official \$249 , retail often \$250–\$560+	Best practical starting point for AI cabin-camera experiments. NVIDIA lists it at \$249, but retailer prices vary.

NVIDIA Jetson Orin NX dev kit / edge device	about \$970–\$1,200+	More power than Orin Nano; useful for heavier models or multiple cameras. NVIDIA's FAQ lists Orin NX module MSRP at \$449–\$699 depending on 8GB/16GB, before carrier board/case/storage.
NVIDIA Jetson AGX Orin Developer Kit	official about \$1,999	High-end prototyping. Overkill for one simple inside camera, but useful for advanced multi-camera AI.
TI SK-TDA4VM / J721EXSKG01EVM	about \$249–\$311	More automotive/embedded-vision style. TI lists the part at \$249, and DigiKey showed related pricing around \$311.25.
NXP i.MX 8M Plus EVK	often quote/distributor dependent	Good embedded vision platform with NPU + dual ISP/camera support, but public price is less consistently shown. NXP describes it as an evaluation kit for i.MX 8M Plus vision/ML applications.
OMNIVISION OAX8000 / OAX4600 DMS ASIC	not normally public	Production-grade DMS/OMS chips. You usually contact OMNIVISION/distributor for pricing; not like buying a Raspberry Pi.

My practical recommendation:

For learning/prototype: **Jetson Orin Nano Super** or **OAK-D**.

For automotive-style development: **TI TDA4VM**.

For very cheap experiments: **Raspberry Pi 5 + camera**, but it is not ideal for real-time DMS AI unless the model is small.

give me the price for Jetson Orin Nano Super
Thought for 7s

The **NVIDIA Jetson Orin Nano Super Developer Kit** is currently around:

\$249 USD from official/major sellers like **Micro Center** and **Seeed Studio**. Some resellers list it much higher, around **\$299–\$560+**, but I would treat **\$249** as the normal target price.

For your car inside-camera prototype, budget extra for:

Item

**Approx.
cost**

Jetson Orin Nano Super Developer Kit	\$249
Camera module, preferably IR/global shutter if doing driver monitoring	\$30–\$150+
Power supply / adapter	\$15–\$30
microSD / NVMe storage	\$15–\$60
Case / cooling / mount	\$20–\$60

So realistically, a usable setup may be around **\$320–\$500**, depending on the camera.

I'll frame it like a real product pipeline: hardware → camera/lighting → image processing/AI → embedded deployment → vehicle integration → testing/safety. This is one of those projects where the “AI model” is only half the battle; the camera and lighting are the sneaky hard parts.

Thought for 9s

To develop a **DMS — Driver Monitoring System**, think of it as this pipeline:

camera → face/eye detection → driver-state estimation → alert/vehicle action → logging/testing

For a prototype, you can build it with **Jetson Orin Nano Super + IR camera**. For production automotive, you'd eventually move toward something like **TI TDA4VM**, **NXP**, or a dedicated DMS ASIC.

1. Decide what your DMS must detect

Start with a small MVP. Don't try to build everything at once.

Good MVP features:

Feature	Meaning
Driver face detected	Is a driver visible?
Eye open / closed	Drowsiness detection

Blink duration	Long eye closure = possible fatigue
Head pose	Looking forward, left, right, down
Gaze direction	Eyes looking road / phone / side
Distraction	Looking away too long
Yawning	Fatigue signal
Phone usage	Optional, harder
Seatbelt / smoking / eating	Optional object detection

For the first version, I'd build only:

face detection + eye state + head pose + drowsiness/distraction alert

That is already enough to make a meaningful DMS.

2. Choose hardware

For a prototype:

Component	Recommendation
Compute board	NVIDIA Jetson Orin Nano Super
Camera	IR camera or RGB-IR camera
Lens	Wide enough to see driver face, usually 60–90° FOV
IR LEDs	Needed for night operation
Storage	microSD or NVMe
OS	Ubuntu + JetPack
AI runtime	TensorRT / ONNX Runtime / OpenCV

Important: for real DMS, a normal RGB webcam is weak because nighttime and sunlight changes are brutal. Use **IR camera + IR illuminator** if possible.

NVIDIA lists the Jetson Orin Nano Super Developer Kit at **\$249**, and TI's TDA4VM is an automotive vision SoC with **8 TOPS** AI acceleration for vision perception/analytics.

3. Camera placement

Inside a car, placement matters a lot.

Common camera positions:

Position	Pros	Cons
Steering column	Best face angle	Harder mechanical mounting
Dashboard center	Easy install	Side angle can hurt gaze accuracy
Rear-view mirror area	Good visibility	More distance, lighting variation
A-pillar	Common in OEM systems	Side angle, occlusion risk

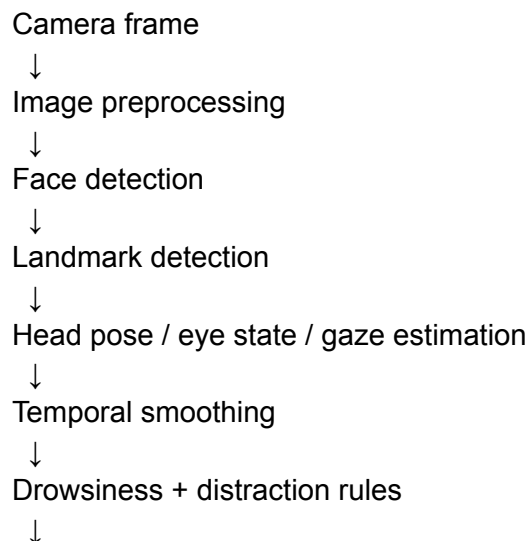
For your prototype, use:

dashboard / steering-column camera facing the driver's face

Try to keep the driver's face around the center of the image.

4. Build the software pipeline

A normal DMS pipeline looks like this:



Alert output

A. Capture camera frames

Use:

- OpenCV
- GStreamer
- Jetson camera stack
- V4L2

For prototype:

```
import cv2
```

```
cap = cv2.VideoCapture(0)
```

```
while True:
```

```
    ok, frame = cap.read()
```

```
    if not ok:
```

```
        break
```

```
    cv2.imshow("DMS Camera", frame)
```

```
    if cv2.waitKey(1) == 27:
```

```
        break
```

```
cap.release()
```

```
cv2.destroyAllWindows()
```

5. Detect face and landmarks

You need face landmarks like:

- eye corners
- eyelids
- nose
- mouth corners
- chin
- face outline

Options:

Tool/model

Good for

MediaPipe Face Mesh	Very fast prototype
OpenCV DNN face detector	Simple face detection
YOLO face model	Robust face detection
Dlib landmarks	Classic, but older/slower
Custom trained model	Production-grade later

For MVP, use **MediaPipe Face Mesh** first. It gives face landmarks quickly and lets you calculate eyes, mouth, and head pose.

6. Drowsiness detection

The classic simple method is **Eye Aspect Ratio**, often called **EAR**.

Concept:

If eyes remain closed for more than N frames/seconds → drowsiness alert

Simple logic:

```
if eye_closed_duration > 1.5 seconds:
    alert = "DROWSY"
```

You should not alert on one blink. Normal blinks are short. Drowsiness is usually based on **long eye closure** or repeated slow blinking.

Better drowsiness signals:

Signal	Meaning
Long eye closure	likely microsleep
High blink frequency	fatigue
Slow eyelid reopening	fatigue
Yawning	fatigue
Head nodding	fatigue
PERCLOS	percentage of time eyes are closed

For a more serious DMS, use **PERCLOS**:

PERCLOS = % of time eyes are mostly closed over a time window

Example:

If eyes closed more than 40% of the last 60 seconds → fatigue risk

7. Distraction detection

Distraction usually means:

Driver is not looking at the road for too long

You can estimate this from:

- head pose: yaw, pitch, roll
- gaze direction
- face not detected
- eyes looking down

Simple rule:

if head_yaw > 30 degrees for more than 2 seconds:
alert = "LOOKING_AWAY"

Examples:

Condition	Possible state
head turned left/right	looking away
head down	looking at phone/lap
face missing	driver not visible / occluded
eyes closed long	drowsy
mouth open long	yawning

8. Add temporal smoothing

This part is super important. Without smoothing, your DMS will be noisy and annoying.

Do not alert from one frame.

Use state over time:

```
looking_away_frames += 1
eye_closed_frames += 1
face_missing_frames += 1
```

Then alert only after thresholds:

```
eye closed > 1.5s
looking away > 2.0s
face missing > 3.0s
```

At 30 FPS:

```
1.5 seconds = 45 frames
2.0 seconds = 60 frames
3.0 seconds = 90 frames
```

9. Output alerts

For prototype:

- show text on screen
- play sound
- write event logs
- send CAN message later
- send HTTP/MQTT event

Example event:

```
{
  "event_type": "DROWSINESS_DETECTED",
  "severity": "HIGH",
  "timestamp": "2026-05-28T12:00:00Z",
  "confidence": 0.91,
  "signals": {
    "eye_closed_duration_ms": 1800,
    "head_pose_yaw_deg": 4.2
  }
}
```

10. Optimize for Jetson

Python/OpenCV is fine at first, but later optimize:

Stage	Prototype	Optimized
Capture	OpenCV	GStreamer
Model	PyTorch / MediaPipe	ONNX / TensorRT
Inference	CPU/GPU	TensorRT FP16/INT8
Pipeline	single thread	multi-thread capture/inference/output
Logging	print	structured event logs

NVIDIA Jetson Orin Nano Super is intended for edge AI prototyping and supports JetPack/TensorRT workflow, so it is a strong board for this kind of prototype.

11. Test in real car conditions

This is where DMS gets difficult.

Test with:

Condition	Why
Daylight	baseline
Night	IR camera needed
Direct sunlight	face can be overexposed
Sunglasses	eye detection becomes harder
Hat/cap	face occlusion
Different driver heights	camera angle changes
Bumpy road	motion blur
Phone looking-down behavior	distraction detection
Driver looking mirrors	should not always alert
Driver shoulder-checking	should not always alert

A DMS must avoid false alarms. If it beeps every time the driver checks a mirror, users will hate it. Classic car-tech gremlin: technically correct, emotionally unbearable.

12. Development roadmap

I'd build it like this:

Phase 1 — PC prototype

Use laptop webcam first.

Build:

- camera preview
- face detection
- landmarks
- eye open/closed
- head pose
- basic alert overlay

Phase 2 — Jetson prototype

Move to Jetson Orin Nano Super.

Add:

- IR camera
- night testing
- FPS measurement
- TensorRT optimization if needed

Phase 3 — Vehicle prototype

Put it in a car.

Add:

- real camera mount
- power from car adapter
- alert speaker
- event logging
- heat/vibration testing

Phase 4 — Production direction

Move toward:

- automotive-grade camera
- automotive SoC or ASIC
- CAN/Ethernet integration
- safety requirements
- privacy/security design
- compliance testing

Euro NCAP has active Safe Driving protocols covering **Occupant Monitoring** and **Driver Engagement**, and its 2026 protocols increase attention on driver/occupant monitoring as part of safety assessment.

Simple MVP architecture

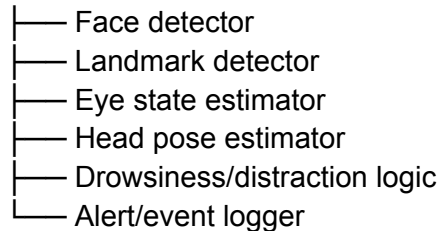
IR Camera



Jetson Orin Nano Super



DMS App



Display / buzzer / dashboard signal

My recommended first stack

For your first working DMS prototype:

Layer	Choice
Board	Jetson Orin Nano Super
Camera	USB IR camera or MIPI CSI IR camera
Language	Python first
Libraries	OpenCV + MediaPipe
AI optimization	Later: ONNX/TensorRT

First features	face detection, eye closure, head pose, alert
Target FPS	20–30 FPS minimum

Once the prototype works, then replace prototype models with more robust trained models and optimize for embedded deployment.

Thought for a couple of seconds

With the **DMS model/pipeline** we discussed — camera + face detection + face landmarks + head pose + eye/mouth analysis — you can detect these driver actions/states:

Main actions/states you can detect

1. Driver present / not present

Detect whether a face is visible in the driver seat.

Examples:

DRIVER_DETECTED
DRIVER_NOT_DETECTED
FACE_OCCLUDED

Useful for checking whether the driver is visible, camera is blocked, or the driver moved out of view.

2. Eye closed / drowsiness

Detect if the driver's eyes are closed for too long.

Examples:

EYES_OPEN
EYES_CLOSED
LONG_EYE_CLOSURE
DROWSINESS_DETECTED
MICROSLEEP_RISK

Simple rule:

if eyes_closed_duration > 1.5 seconds:
DROWSINESS_DETECTED

This is one of the most important DMS features.

3. Blinking behavior

Detect normal blink, long blink, frequent blink.

Examples:

NORMAL_BLINK
LONG_BLINK
FREQUENT_BLINKING
SLOW_EYE_REOPENING

This helps estimate fatigue.

4. Looking away

Detect if the driver is not looking forward.

Examples:

LOOKING_FORWARD
LOOKING_LEFT
LOOKING_RIGHT
LOOKING_DOWN
LOOKING_UP
LOOKING_AWAY

This can be based on **head pose** and sometimes **gaze direction**.

Simple rule:

if looking_away_duration > 2 seconds:
DISTRACTION_DETECTED

5. Looking down at phone / lap

This is a special case of distraction.

Examples:

LOOKING_DOWN
PHONE_DISTRACTION_RISK
LAP_DISTRACTION_RISK

Important note: with only face landmarks, you can detect **looking down**, but you cannot always prove the driver is looking at a phone unless you also add **object detection** for phone.

6. Head pose

Detect head direction using face landmarks.

Examples:

HEAD_FORWARD
HEAD_TURNED_LEFT
HEAD_TURNED_RIGHT
HEAD_TILTED
HEAD_DOWN
HEAD_UP

Head pose gives you:

Signal	Meaning
Yaw	left/right head turn
Pitch	up/down
Roll	head tilt

7. Yawning

Detect if the mouth is open for a long time.

Examples:

MOUTH_CLOSED
MOUTH_OPEN
YAWNING_DETECTED

FATIGUE_RISK

Simple rule:

if mouth_open_duration > 1.5 seconds:
 YAWNING_DETECTED

But be careful: talking/singing can look similar to yawning, so you need time-based logic.

8. Face occlusion

Detect when part of the face is blocked.

Examples:

FACE_PARTIALLY_OCCLUDED
EYES_NOT_VISIBLE
MOUTH_NOT_VISIBLE
CAMERA_BLOCKED

Common cases:

Case	Detection difficulty
sunglasses	medium/hard
mask	medium
hand on face	medium
hat/cap	medium
strong sunlight	hard
camera covered	easy

Extra actions if you add object detection

The basic face/landmark model does **not** reliably detect objects. But if you add a YOLO/object-detection model, you can detect more driver behaviors.

Action	Need extra object model?
Holding phone	Yes
Smoking	Yes
Eating	Yes
Drinking	Yes
Seatbelt detection	Yes
Hands off wheel	Yes, usually
Child/passenger detection	Yes
Driver identity recognition	Separate face recognition model

Examples:

PHONE_IN_HAND
 SMOKING_DETECTED
 DRINKING_DETECTED
 EATING_DETECTED
 SEATBELT_NOT_FASTENED
 HANDS_OFF_WHEEL

Good MVP detection list

For your first DMS prototype, I recommend detecting only these:

DRIVER_DETECTED
 FACE_NOT_DETECTED
 EYES_CLOSED
 DROWSINESS_DETECTED
 LOOKING_FORWARD
 LOOKING_LEFT
 LOOKING_RIGHT
 LOOKING_DOWN
 DISTRACTION_DETECTED
 YAWNING_DETECTED

That is enough for a useful first version.

Detection levels

You can organize detected actions like this:

Level 1: Raw signals

- face detected
- eyes open/closed
- mouth open/closed
- head yaw/pitch/roll

Level 2: Driver states

- looking away
- drowsy
- yawning
- distracted
- face missing

Level 3: Alerts

- warning
- high-risk warning
- emergency attention alert

Example output:

```
{
  "driver_state": "DISTRACTED",
  "face_detected": true,
  "eye_state": "OPEN",
  "head_pose": {
    "yaw": 38.5,
    "pitch": 4.2,
    "roll": 1.1
  },
  "looking_away_duration_ms": 2400,
  "alert_level": "WARNING"
}
```

So, with the basic DMS model, you mainly detect **face, eyes, mouth, head direction, drowsiness, yawning, and looking-away distraction.**

For **phone, smoking, eating, drinking, seatbelt, or hands-on-wheel**, you need an additional **object/action detection model.**

I'll separate this into **ready-to-use software SDK/model products** and **hardware modules/SoCs that already include DMS capability**, because vendors package "completed model" in both ways.

Thought for 6s

Yes. There are **completed DMS products/models**, but most serious ones are **commercial SDKs or integrated hardware+software products**, not free downloadable `.pt` / `.onnx` files.

1. Smart Eye DMS / AIS

Smart Eye provides automotive-grade DMS software for driver attention, gaze, drowsiness, distraction, and interior sensing. They say their DMS software has been selected by **24 OEMs for 372 car models**.

They also have **AIS**, which is closer to a plug-and-play aftermarket/fleet product. Smart Eye describes AIS as an end-to-end driver monitoring system for fleet and aftermarket vehicles, detecting drowsiness, distraction, and more.

Best for: **OEM, fleet, aftermarket DMS product**

Not best for: small developer wanting a simple open-source model file.

2. Seeing Machines Guardian

Seeing Machines Guardian is a completed DMS product used in fleet/commercial vehicles. It monitors drowsiness and distraction in real time. Their product page says Guardian tracks driver eyes for distraction analysis and detects early drowsiness.

Best for: **fleet vehicle safety product**

Typical form: camera/device + cloud/fleet system, not just a model.

3. Cipia Driver Sense / Cabin Sense / FS10

Cipia Driver Sense is a commercial DMS software product. It detects head pose, gaze direction, driver position, drowsiness, distraction, phone-holding, and more.

Cipia also has **Cabin Sense** for driver + occupant monitoring, and **Cipia-FS10**, which is a video-telematics/DMS product for fleets.

One note: HARMAN announced acquisition of Cipia assets, so the product may now be handled through HARMAN/Cipia channels depending on the exact offering.

Best for: **OEM integration, fleet monitoring, production-style DMS**

4. Mobileye DMS

Mobileye announced its own **Mobileye Driver Monitoring System**, integrated with Mobileye ADAS on a single chip. This is more for automakers, not hobby development.

Best for: **automotive OEM ADAS platform**

5. OMNIVISION + Smart Eye interior sensing solution

There are also complete **camera/processor + DMS software** solutions. For example, Smart Eye and OMNIVISION announced an automotive interior sensing solution targeting GSR and Euro NCAP requirements.

Best for: **production automotive camera module direction**

Simple answer

If you want a **completed commercial DMS product**, look at:

Product	Type	Detects
Smart Eye DMS / AIS	Software / aftermarket system	gaze, attention, drowsiness, distraction
Seeing Machines Guardian	Fleet DMS product	drowsiness, distraction, eye attention
Cipia Driver Sense	DMS software	head pose, gaze, drowsiness, distraction, phone holding
Cipia FS10	Fleet camera product	DMS + video telematics
Mobileye DMS	OEM ADAS platform	driver monitoring integrated with ADAS

For your case — if you want to build your own device with **Jetson Orin Nano Super** — the easiest route is probably:

prototype with open models first, then later license a commercial SDK if you need production-grade accuracy.

For example:

Jetson Orin Nano Super

+ IR camera

+ OpenCV / MediaPipe / YOLO prototype

+ later commercial DMS SDK if needed

The commercial products are usually not sold like “download model for \$99.” You usually contact sales, describe hardware/volume/use case, and they provide SDK, evaluation kit, or full device pricing.

I want to buy completed product for Driver Monitoring System, Occupant Monitoring System

I'll focus on **buyable completed systems**, not chips or development boards. For DMS/OMS there are two very different markets: aftermarket fleet devices you can buy/install, and OEM-grade systems that require vendor sales/integration.

Thought for 15s

Yes — you can buy completed products, but you should choose based on your target:

For your own fleet / commercial vehicles: buy an aftermarket DMS camera device.

For building a car/OEM product: contact vendors for an integrated DMS/OMS solution or SDK.

AI based Driver Monitoring System CIPIA (RS232)

\$370.00•Cipia-based driver monitoring with RS232 interface.



FleetFlix AI+ Pro Dual Dash AI Camera
\$390.00•Fleet AI+ dual dash camera with fleet DMS features.



Driver•i D-215
\$399.00•Fleet dashcam with AI and driver monitoring capabilities.



JC450 LTE Dashcam System with ADAS, DMS, and Cloud Storage
\$270.67•LTE dashcam with ADAS, DMS, and cloud storage.



Blackvue DR770X BOX-3CH DMS Truck Pro
\$614.99•Pro-grade truck dashcam with DMS and advanced features.



LDS MDVR AI Driver Fatigue Monitoring DMS Camera
\$144.96•High-end truck DMS with premium features.



AI Driver Monitoring System — Commercial Grade
\$119.99•Affordable driver fatigue monitoring with AI.



DMS Anti Fatigue Driver Monitoring System AI Dash Camera
\$186.31•Low-cost dash camera for driver monitoring.



DMS Driver Monitoring System AI Fatigue Warning Device
\$63.98•Budget-friendly driver fatigue warning device.

Completed DMS products you can buy

1. Cipia-FS10 / FS10 Plus

This is one of the better-known completed aftermarket DMS products. Cipia-FS10 is described as an in-cabin video telematics and driver monitoring product for fleets, using computer vision and AI to monitor driver state and provide real-time alerts.

A distributor listing shows a **Cipia RS232 DMS device around \$370**. The FS10 Plus is described as detecting driver state, drowsiness, distraction, head pose, eyelids, gaze direction, phone use, smoking, seatbelt, and identity recognition.

Good choice if: you want a serious aftermarket DMS device for commercial vehicle/fleet use.

2. Seeing Machines Guardian

Guardian is a completed fleet DMS product. Seeing Machines says it continuously monitors driver drowsiness and tracks eye attention/distraction in real time.

Good choice if: you want a professional fleet-grade system and are okay contacting sales/distributors.

Price: usually not public; likely quote-based.

3. Smart Eye AIS / AIS+

Smart Eye AIS is a complete aftermarket driver support system for fleets and small-volume vehicle manufacturers. It detects drowsiness and distraction, and Smart Eye describes it as hardware + software for aftermarket installation.

Smart Eye launched **AIS+** with real-time haptic alerts and optional video recording.

Good choice if: you want a high-end commercial DMS system.

Price: usually quote-based.

Completed DMS + dashcam products

These are easier to buy online, but they may be less “OEM-grade” than Cippa, Smart Eye, or Seeing Machines.

Product	Approx. online price	Notes
FleetFlix AI+ Pro Dual Dash AI Camera	\$390	AI DMS, distracted driving, drowsiness, smartphone usage.
Driver-i D-215	\$399	Fleet dashcam with driver alerts/analysis.
JC450 LTE Dashcam with ADAS + DMS	\$270	Multi-channel dashcam, ADAS, DMS, cloud.
BlackVue DR770X BOX-3CH DMS Truck Pro	\$615	Higher-end dashcam with DMS camera.
LDS FHD831 DMS Camera	\$145	DMS fatigue/distraction camera module.
Budget AI DMS devices	\$60–\$190	Cheap, but quality/accuracy may be much lower.

What about OMS?

For **Occupant Monitoring System**, it is harder to buy a simple retail product. OMS is usually OEM/integration-oriented.

OMS detects things like:

- passenger presence
- child left behind
- seat occupancy
- seatbelt state
- object/cargo vs human
- occupant posture
- airbag-related occupant classification

For OMS, look more at:

Vendor	Type
Smart Eye	Interior sensing / DMS + OMS solution
Seeing Machines	Driver and occupant monitoring technology

indie Semiconductor DOMS

Combined DMS + OMS software for OEM/Tier-1

OMNIVISION

Camera/sensor/processor solutions for DMS/OMS

NXP / TI / Analog Devices ecosystem

Hardware platform for DMS/OMS integration

indie Semiconductor describes its DOMS software as combined **driver and occupant monitoring using a single camera**, aimed at OEMs and Tier-1s. OMNIVISION also describes OMS features such as distinguishing cargo from human occupants and detecting objects left behind.

My recommendation

For **buying now and testing in a vehicle**, start with:

Best serious aftermarket choice:
Cipia-FS10 / FS10 Plus

Best fleet-grade but quote-based:
Seeing Machines Guardian or Smart Eye AIS

Cheaper test product:
FleetFlix AI+ Pro or JC450 ADAS + DMS

For **real DMS + OMS product development**, contact **Smart Eye, Seeing Machines, Cipia/HARMAN, indie Semiconductor, or OMNIVISION**. Retail DMS cameras are easy to buy; true OMS is usually not a simple Amazon-style product.

give me the product catalog for Cipia-FS10 / FS10 Plus

I'll pull together a compact catalog-style view: product family, key features, detected events, interfaces, and buying/contact path.

Thought for 10s

Below is a **catalog-style summary** for **Cipia-FS10 / Cipia-FS10 Plus**.

Cipia-FS10 / FS10 Plus Product Catalog

1. Product name

Cipia-FS10 / Cipia-FS10 Plus

This is an **aftermarket in-cabin video telematics + Driver Monitoring System device** for fleets. It is designed to be installed inside vehicles and monitor the driver in real time using AI/computer vision.

2. Main use case

Fleet driver safety monitoring

It is mainly used for:

- trucks
- buses
- commercial vehicles
- mining vehicles
- logistics fleets
- taxi / ride-hailing fleets
- company vehicle fleets

The purpose is to detect risky driver behavior and send alerts to the driver and/or fleet manager.

3. Core DMS detection features

Cipia-FS10 Plus can detect:

Detection	Meaning
Drowsiness	Driver appears sleepy or falling asleep

Distraction	Driver is not paying attention
Seatbelt usage	Detects improper seatbelt usage
Smoking	Detects driver smoking
Phone holding / phone use	Detects holding a phone while driving
Driver recognition	Recognizes enrolled drivers
Camera status / tampering	Can detect camera status or misuse depending on integration

Cipia says the in-cabin sensor enables detection of **driver drowsiness, distraction, seatbelt usage, smoking, and holding a phone while driving**, with real-time driver alerts. It also supports driver recognition for enrolled drivers.

4. FS10 vs FS10 Plus

Cipia-FS10

Base DMS/video-telematics device.

Typical functions:

- in-cabin driver monitoring
- IR-based driver sensing
- driver alerts
- fleet manager alerts
- integration with fleet management systems
- detection of fatigue, distraction, smoking, phone holding, and seatbelt misuse

Cipia-FS10 Plus

Newer / enhanced version.

Adds or improves:

- optional **road-facing camera**
- dashcam video recording
- road-facing snapshots or footage during DMS events
- correlation with external ADAS events
- hybrid communication mode: cellular immediately or Wi-Fi upload later
- improved fleet event analysis

Simple way:

FS10 = in-cabin DMS device

FS10 Plus = FS10 + optional road-facing camera + stronger video telematics features

5. Camera / sensor system

The FS10 uses an **in-cabin IR sensor/camera** for driver monitoring. IR is important because DMS must work at night and in low light.

FS10 Plus can also support an optional **road-facing camera** for dashcam footage and event reconstruction.

6. Alerts

The system can provide:

Alert target	Description
Driver alert	Real-time alert inside the vehicle

Fleet manager alert	Event/notification sent to fleet system
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Event footage	Video/snapshot around the event
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Post-incident analysis	Helps review what happened before/after incident
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Cipia describes the device as providing real-time audio/visual alerts to the driver and tailored alerts to the fleet manager or dispatcher.

7. Fleet/cloud integration

Cipia-FS10 Plus can work with:

- existing fleet management server
- FSCloud / cloud dashboard depending on deployment
- telematics gateway/platform integrations
- API-style fleet system integrations through partners

One known integration is with **DCT Pegasus IoT platform** using the **Syrus 4G IoT Gateway**.

8. Communication / connectivity

Available information mentions:

Feature	Description
LTE / 4G option	Some FS10 Plus listings mention LTE 4G version

Wi-Fi upload	Hybrid mode can store videos/snapshots locally and upload later
Cellular upload	Events/images/videos can be sent immediately
Fleet server integration	Sends events to fleet management system

FS10 Plus has a hybrid communication mode where videos and snapshots can be sent immediately over cellular or stored locally until secure Wi-Fi is available. Cipia said this can reduce data usage by up to 90%.

9. Road-facing / ADAS capability

FS10 Plus can use an optional road-facing camera.

This enables:

- dashcam video recording
- road-facing snapshots/footage during DMS events
- correlation with external ADAS events
- better accident reconstruction

Cipia also announced newer ADAS capabilities for FS10 in 2025, including a dual-camera road-and-driver monitoring approach.

10. Approximate price

Public pricing is limited, but one distributor listing shows:

Product/listing	Approx. price
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AI based Driver Monitoring System CIPIA RS232 \$370 USD

That listing describes it as a Cipia-FS10 DMS device for driver fatigue and distraction detection.

Other sellers may use quote-based pricing, especially for fleet volume, LTE plans, cloud subscription, installation, and integration.

11. Buying/contact path

There are a few ways to buy or inquire:

Path	Notes
Cipia / FS10 support	HARMAN says the FS10 aftermarket program was formerly managed by Cipia and is not owned by HARMAN ; they list FScLOUD_support@cipia.com for FS10 program information.
FleetSafe Australia	Has FS10 Plus overview and product information.
Digital Communications Technologies / DCT	Partner/integration route through Pegasus IoT/Syrus gateway.
TrackingHardware UK	Lists Cipia FS-10 and references FS10 Plus datasheet.
Anstel Global	Shows a Cipia RS232 DMS device listing at \$370.

12. Important limitation: DMS, not full OMS

Cipia-FS10 / FS10 Plus is mainly a **Driver Monitoring System / fleet video telematics product**.

It is not the same as a full **Occupant Monitoring System** for all passengers, child presence, passenger posture, airbag classification, etc. For OMS, you would look more at Cipia/HARMAN **Cabin Sense**, Smart Eye interior sensing, Seeing Machines, indie Semiconductor DOMS, or OMNIVISION-based solutions.

Best summary

Cipia-FS10 Plus is a completed aftermarket **AI DMS product** for fleets. It detects **drowsiness, distraction, seatbelt misuse, smoking, phone holding, and driver identity**, and it can optionally add a **road-facing camera** for dashcam recording and event reconstruction. Public single-device pricing may be around **\$370**, but serious fleet pricing is usually quote-based.